

数据库第一次作业

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2351136

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No.
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- 2.6 解: a. $\Pi_{\text{person-name}} (\sigma_{\text{city} = \text{"Miami"}} (\text{employee}))$
 b. $\Pi_{\text{person-name}} (\sigma_{\text{salary} > 10000} (\text{works}))$
 c. $\Pi_{\text{person-name}} ((\sigma_{\text{city} = \text{"Miami"}} (\text{employee})) \bowtie (\sigma_{\text{salary} > 10000} (\text{works})))$

2.7 解:

- a. $\Pi_{\text{branch-name}} (\sigma_{\text{branch-city} = \text{"Chicago"}} (\text{branch}))$
 b. $\Pi_{\text{ID}} (\text{Cborrower}) \bowtie \sigma_{\text{branch-name} = \text{"Downtown"}} (\text{loan})$

2.8 解:

- a. $\Pi_{\text{ID}, \text{person-name}} (\sigma_{\text{company-name} \neq \text{"BigBank"}} (\text{works}))$
 b. $\Pi_{\text{ID}, \text{person-name}} (\text{works}) - \Pi_{\text{ID}, \text{person-name}} (P_{E_1} (\text{works}) \bowtie P_{E_2} (\text{works}))$
 $E_1: \text{salary} < E_2: \text{salary}$

2.14 解:

- a. $\Pi_{\text{ID}, \text{person-name}} (\sigma_{\text{company-name} = \text{"BigBank"}} (\text{works}))$
 b. $\Pi_{\text{ID}, \text{person-name}, \text{city}} (\sigma_{\text{company-name} = \text{"BigBank"}} (\text{works})) \bowtie \sigma_{\text{company-name} = \text{"BigBank"}} (\text{company})$
 c. $\Pi_{\text{ID}, \text{person-name}, \text{street}, \text{city}} (\sigma_{\text{company-name} = \text{"BigBank"} \wedge \text{salary} > 10000} (\text{works})) \bowtie \sigma (\text{employee})$

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- d. $\Pi_{\text{ID}, \text{person-name}} (\text{employee} \bowtie \text{company})$
 $\text{employee.city} = \text{company.city}$

2.15.

- 解: a. $\Pi_{\text{loan-number}} (\sigma_{\text{amount} > 10000} (\text{loan}))$
 b. $\Pi_{\text{ID}} (\sigma_{\text{balance} > 6000} (\text{account}) \bowtie \text{depositor})$
 c. $\Pi_{\text{ID}} (\sigma_{\text{branch-name} = \text{"Uptown"} \wedge \text{balance} > 6000} (\text{account}) \bowtie \text{depositor})$